Comparative Analysis of CNN-Based and Vision Transformer-Based Image Caption Generator Using Transformer Models

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***Abstract*—Image captioning combines visual recognition with autoregressive language generation. This paper compares ResNet-50 and ViT-B/16 encoders using the same two-layer Transformer decoder and spatial memory: 49 projected CNN grid tokens or 196 projected ViT patch tokens. On the 1,000-image Flickr30k Karpathy test split, ViT has higher metric point estimates, including BLEU-4 (0.2251 vs. 0.2098) and CIDEr (0.4424 vs. 0.4192). A paired percentile bootstrap over test images found that the unadjusted 95% intervals for ViT-minus-CNN differences excluded zero for BLEU-3 and BLEU-4, but not the other five metrics. ResNet trained in 188.5 minutes over 22 epochs, compared with 492.27 minutes for ViT; mean greedy-decoding latency was 36.01 ms/image for ResNet and 37.55 ms/image for ViT. Since each architecture was trained with one seed, these intervals quantify test-image sampling uncertainty only and not training-run variability. The results show a quality–compute trade-off for this dataset and implementation, not a universal encoder ranking.**

***Index Terms*—image captioning, ResNet-50, Vision Transformer, ViT-B/16, Transformer decoder, Flickr30k, BLEU, CIDEr, encoder-decoder, multimodal learning**

I. Introduction

Image captioning is a challenging multimodal task that aims to automatically generate natural language descriptions for images by combining techniques from Computer Vision (CV) and Natural Language Processing (NLP). Unlike traditional image classification or object detection, image captioning requires a system to understand not only the objects present in an image but also their relationships, attributes, and contextual information in order to produce coherent and grammatically correct sentences. The ability to automatically describe visual scenes has become increasingly important due to its applications in assistive technologies for visually impaired individuals, intelligent image retrieval, autonomous vehicles, surveillance systems, robotics, social media content analysis, and human-computer interaction.

Early image captioning systems relied on handcrafted visual features combined with template-based or retrieval-based sentence generation. The rapid advancement of deep learning introduced encoder-decoder architectures inspired by neural machine translation, in which Convolutional Neural Networks (CNNs) encode visual representations and Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, generate captions sequentially.

Despite remarkable improvements, RNN-based decoders suffer from difficulties in capturing long-range dependencies and cannot be fully parallelized during training. The Transformer architecture addressed these limitations by replacing recurrence with self-attention mechanisms that enable efficient modeling of global dependencies. Transformer-based decoders have since become the dominant approach in modern image captioning systems.

CNN architectures such as ResNet remain highly effective at learning hierarchical local features and are widely adopted as image encoders due to strong inductive biases for translation equivariance and locality. Vision Transformers (ViTs) have emerged as a powerful alternative by applying self-attention directly to sequences of image patches, enabling explicit modeling of long-range spatial dependencies. While ViTs have demonstrated state-of-the-art performance across many vision tasks, they generally require larger datasets and more computational resources than CNN-based models.

This study compares pretrained ResNet-50 and ViT-B/16 encoders paired with an identical Transformer decoder on Flickr30k. Unlike a global-vector conditioning design, the decoder cross-attends to a sequence of projected spatial CNN or ViT patch features. We report standard automatic caption metrics, training time, validation-loss curves, model parameter counts, and greedy-decoding latency. The comparison is an empirical study of these encoder and tokenization choices under one training protocol; it does not introduce a new captioning architecture or establish performance beyond Flickr30k.

The major contributions of this paper are:

- A controlled implementation in which ResNet-50 grid tokens and ViT-B/16 patch tokens are supplied to the same Transformer decoder.
- Evaluation on the 1,000-image Flickr30k Karpathy test split using BLEU-1–4, METEOR, ROUGE-L, and CIDEr.

- Reporting of convergence, parameter counts, final training time, and greedy-decoding latency, with qualitative examples of both correct and erroneous captions.

II. Literature Review

Image captioning integrates computer vision and natural language processing to generate textual descriptions of visual content. Recent surveys review the transition from recurrent encoder–decoder systems to Transformer and vision–language models [1], [2].

Vinyals *et al.* introduced the Neural Image Caption (NIC) model, establishing the encoder-decoder paradigm by coupling a pretrained GoogLeNet encoder with an LSTM decoder [3]. Xu *et al.* extended this framework with soft attention, enabling the decoder to focus on spatially relevant image regions at each generation step [4]. Anderson *et al.* further improved caption quality by incorporating bottom-up attention over Faster R-CNN object proposals, achieving state-of-the-art performance on MS COCO [5].

Vaswani *et al.* introduced the Transformer, replacing sequential recurrence with multi-head self-attention to enable fully parallelized training and better modeling of long-range dependencies [6]. Transformer decoders rapidly superseded LSTM-based generators in image captioning by producing more fluent and contextually consistent captions. He *et al.* introduced ResNet, addressing the degradation problem in very deep CNNs via residual skip connections, making ResNet-50 one of the most widely adopted visual encoders for captioning tasks [7].

A major paradigm shift occurred with the Vision Transformer (ViT), which partitions images into fixed-size patches and applies self-attention across the patch sequence [8]. Captioning work has since explored full Transformer image encoders, including CPTR, and language-model conditioning approaches such as ClipCap [9], [10]. Caption-specialized Transformer refinements include RSTNet and dual-level collaborative attention, while VinVL and later work scale vision–language pretraining or combine global and local visual features [11]–[16]. These systems are not directly comparable to our Flickr30k experiment because their datasets, pretraining, decoders, and training protocols differ. Grid-token representations have also been studied in adjacent vision-language tasks; for example, Jiang *et al.* analyze grid features for visual question answering rather than caption generation [17].

Large-scale vision–language pretraining has also advanced caption generation. BLIP uses filtered image–text data for unified understanding and generation, while BLIP-2 connects frozen image encoders and language models through a lightweight Querying Transformer [18], [19]. ChatCaptioner uses iterative question answering to enrich descriptions [20]. Automatic overlap metrics do not directly measure object factuality; POPE provides a benchmark for object hallucination in vision–language models [21]. These works motivate factuality analysis but are not baselines in the present experiment.

The present study addresses a narrower question than comparisons with large pretrained captioning systems: how two ImageNet-pretrained encoders perform when their spatial token sequences are projected to a common dimension and supplied to an otherwise shared decoder. This setup permits a controlled comparison of ResNet grid features and ViT patch features, but a single dataset and seed do not establish generalization or statistical significance.

III. Methodology

Both captioning systems adopt an encoder-decoder framework in which a visual encoder extracts image representations and a shared Transformer decoder generates captions autoregressively. The two systems differ solely in the visual encoder; all other components are identical.

A. CNN-Based Captioning Model

The CNN encoder is an ImageNet-pretrained ResNet-50 [7]. We remove its global average-pooling and classification layers and retain the final 7×7 feature grid with 2,048 channels. A learned 1×1 convolution projects each grid location to 512 dimensions, and a learned 7×7 2-D position embedding is added. Flattening the grid yields 49 visual memory tokens of dimension 512 for decoder cross-attention.

B. Vision Transformer-Based Captioning Model

The ViT encoder is an ImageNet-pretrained ViT-B/16 [8]. Each 224×224 image is divided into $14 \times 14 = 196$ non-overlapping 16×16 patches. The patch embeddings and CLS token are processed by 12 Transformer encoder layers with hidden dimension 768 and 12 attention heads. After encoding, the CLS token is excluded from decoder memory; each of the 196 patch tokens is projected from 768 to 512 dimensions by a shared linear layer. The pretrained ViT positional embeddings retain patch-location information.

C. Shared Transformer Decoder

Both models use an identical caption decoder comprising:

- A learned token embedding matrix of shape $|\mathcal{V}| \times 512$, where $|\mathcal{V}| = 2{,}004$.
- A learned positional encoding of shape 128×512.
- A two-layer Transformer decoder with 8 attention heads and model dimension $d_{\text{model}} = 512$.
- A linear output projection from 512 to $|\mathcal{V}|$.

The decoder is trained with a causal mask and cross-attends to either the 49-token ResNet memory or the 196-token ViT memory. Greedy decoding is used at inference with a maximum generation length of 50 tokens.

D. Parameter Counts and Trainable-Layer Status

Table I reports architecture-level parameter counts and trainable-layer status for the final systems. Counts are obtained from the instantiated PyTorch modules used in training.

Both encoders are initialized with ImageNet-pretrained weights and fully fine-tuned. No encoder blocks are frozen. The CNN projection count includes the learned 2-D position embedding; the ViT backbone count excludes the removed classification head.

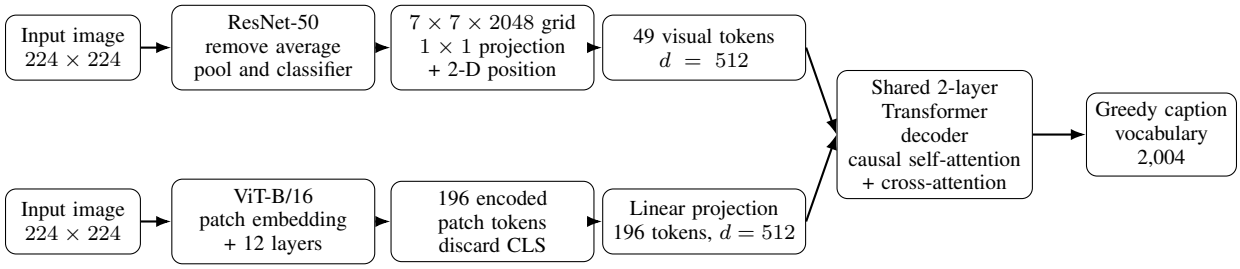


Fig. 1. Compared encoder–decoder systems. Each decoder cross-attends to a sequence of projected visual tokens rather than a single pooled vector.

TABLE I
PARAMETER COUNTS AND TRAINABLE STATUS IN FINAL TRAINING

Component	ResNet System	ViT System
Encoder backbone params	23,508,032	85,798,656
Encoder projection params	1,074,176	393,728
Shared decoder params	10,527,700	10,527,700
Total params	35,109,908	96,720,084
Trainable encoder layers	All layers	All layers
Trainable decoder layers	All layers	All layers

IV. EXPERIMENTAL SETUP

A. Dataset

Both models are trained and evaluated on Flickr30k [22], which contains 31,783 images. The supplied Karpathy split uses 31,014 of these images: 29,000 training images, 1,014 validation images, and 1,000 test images. A programmatic audit confirmed that each notebook's split names matched the supplied JSON, that the splits had no overlap, and that test images had image and caption files.

B. Vocabulary

A shared vocabulary is constructed from the training captions using word-level tokenization. The 2,000 most frequent words are retained, yielding a vocabulary of 2,004 tokens including four special tokens: `<PAD>` (index 0), `<UNK>` (index 1), `<SOS>` (index 2), and `<EOS>` (index 3).

C. Image Preprocessing

Each image is loaded in RGB mode, resized to 224×224, converted to tensor, and normalized using ImageNet statistics (mean $[0.485, 0.456, 0.406]$, standard deviation $[0.229, 0.224, 0.225]$). No geometric or color augmentation is used. For text, captions are lowercased and tokenized using NLTK `word_tokenize`. Sequences are wrapped with `<SOS>` and `<EOS>`, mapped with the shared vocabulary, and padded dynamically per mini-batch using `<PAD>` for loss masking.

D. Data Loader and Batching Details

Training, validation, and test data loaders use batch size 16 with `shuffle=True` for training and `shuffle=False` for validation/test. A custom `collate_fn` stacks images and keeps raw caption lists for on-the-fly token processing. Because `num_workers`, `pin_memory`, and `persistent_workers` are not explicitly set, PyTorch defaults are used (single-process loading and no pinned host memory), so measured epoch times include host-side data loading overhead.

E. Training Configuration

Both models use Adam with differential learning rates and weight decay selected by Optuna [23]. Each search used 15 trials of two epochs and searched dropout, encoder learning rate, projection learning rate, decoder learning rate, and weight decay. The selected ResNet settings were dropout $= 0.1386$, encoder LR $= 1.5291 \times 10^{-5}$, projection LR $= 2.63296 \times 10^{-4}$, decoder LR $= 3.09012 \times 10^{-4}$, and weight decay $= 4.39470 \times 10^{-6}$. The selected ViT settings were dropout $= 0.0488$, encoder LR $= 6.9389 \times 10^{-6}$, projection LR $= 5.4120 \times 10^{-5}$, decoder LR $= 4.05780 \times 10^{-4}$, and weight decay $= 1.08423 \times 10^{-6}$. Cross-entropy loss uses `ignore_index=0` for padding. Final training uses batch size 16, at most 100 epochs, early-stopping patience 5, and minimum validation-loss improvement 10^{-4}. Both final runs use seed 42. The Adam optimizer was used in both final runs; the decoder feed-forward dimension is 2,048 (PyTorch default). The runs used Python 3 and PyTorch 2 in Kaggle Tesla T4 CUDA environments; minor package versions differed between CNN and ViT sessions. The scripts use one default CUDA device and do not use data parallelism.

F. Random Seeds and Determinism

The training scripts seed Python `random`, NumPy, PyTorch CPU, and CUDA generators with 42; Optuna uses `TPESampler(seed=42)`. Fully deterministic CuDNN/CUDA flags were not enabled. The search used the same 15-trial, two-epoch budget for both models. The short trial horizon may not predict performance at full training length, particularly for the ViT's smaller encoder learning rate. Only one final seed was run per architecture, so the results do not estimate run-to-run variability.

G. Timing Protocol and Hardware Utilization Reporting

Reported final-training times exclude Optuna tuning and include the training loop through early stopping. Epoch time includes training, validation, and data-loading/preprocessing overhead. The reported effective throughput divides 29,000 training images by this full epoch time and is therefore an implementation-level proxy, not isolated training throughput.

Greedy-decoding latency is averaged over 1,000 test images. Peak GPU memory was not recorded.

V. Evaluation Metrics

Caption quality is measured on the 1,000-image test set with the `pycocoevalcap` BLEU, METEOR, ROUGE, and CIDEr scorers [24]. Captions are lowercased and tokenized with NLTK `word_tokenize`; tokenized reference strings and generated vocabulary tokens are passed to the scorers. This implementation and tokenization should be considered when comparing scores from other toolkits.

Reference-free image–text measures such as CLIPScore can provide complementary evidence, but are not computed in this study [25].

BLEU (Bilingual Evaluation Understudy) measures n-gram precision between generated and reference captions with a brevity penalty. We report BLEU-1 through BLEU-4 [26].

METEOR (Metric for Evaluation of Translation with Explicit ORdering) computes an F-score based on unigram alignment between hypothesis and reference, incorporating stemming and synonym matching [27].

ROUGE-L (Recall-Oriented Understudy for Gisting Evaluation – Longest Common Subsequence) measures longest-common-subsequence overlap between generated and reference captions, combining sequence-level precision and recall in an F-score [28].

CIDEr (Consensus-based Image Description Evaluation) computes TF-IDF-weighted n-gram similarity between a generated caption and the set of reference captions, rewarding descriptions that are both accurate and consensus-aligned [29].

Inference latency is measured as the average wall-clock time per image on the test set using greedy decoding.

VI. Experimental Results

A. Caption Quality

Table II reports scores on the 1,000 Flickr30k test images. The ViT point estimates are higher on all seven metrics. To quantify test-set sampling uncertainty, we performed 1,000 paired bootstrap resamples using NumPy seed 42: each replicate samples 1,000 image IDs with replacement and uses the same IDs for both systems. Table III reports percentile 95% confidence intervals for the score difference (ViT minus CNN). The unadjusted intervals exclude zero for BLEU-3 and BLEU-4 only. This paired test-set bootstrap does not account for training randomness because only one checkpoint per architecture was evaluated; the intervals are exploratory and no correction for testing seven metrics was applied.

B. Computational Efficiency

Table IV summarizes the computational characteristics of both models. ResNet-50 trained and decoded faster in this run; ViT-B/16 had more parameters and higher caption metric point estimates. Training times exclude Optuna search.

*Effective throughput is training-image count divided by epoch time, which includes validation and data loading.

TABLE II
Caption quality comparison on Flickr30k test set (1,000 images)

Metric	ResNet-50	ViT-B/16
BLEU-1	0.6441	**0.6482**
BLEU-2	0.4373	**0.4505**
BLEU-3	0.3008	**0.3173**
BLEU-4	0.2098	**0.2251**
METEOR	0.1954	**0.1992**
ROUGE-L	0.4886	**0.4964**
CIDEr	0.4192	**0.4424**

TABLE III
Paired bootstrap intervals for metric differences (ViT minus CNN)

Metric	Difference	95% CI
BLEU-1	+0.0041	[−0.0125, +0.0207]
BLEU-2	+0.0132	[−0.0027, +0.0278]
BLEU-3	+0.0165	[+0.0023, +0.0294]
BLEU-4	+0.0152	[+0.0022, +0.0278]
METEOR	+0.0038	[−0.0018, +0.0090]
ROUGE-L	+0.0078	[−0.0009, +0.0165]
CIDEr	+0.0232	[−0.0058, +0.0540]

C. Training Convergence

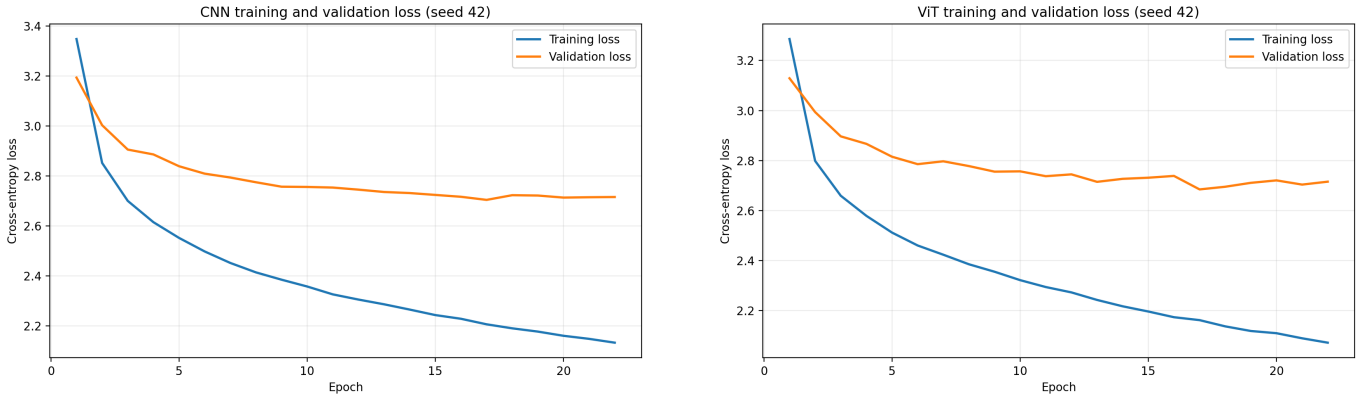
Both models ran 22 epochs. ResNet-50's best validation loss was 2.7038 at epoch 17; ViT-B/16's was 2.6849 at epoch 17. Figure 2 shows that training loss continued to decline after validation loss began to level off, indicating a widening train–validation gap late in training. The best checkpoint was selected by validation loss and restored before test evaluation.

D. Qualitative Examples

Figure 3 illustrates complementary errors on held-out images. For the running dog, the CNN hallucinates a ball, whereas ViT describes the dog and grass in agreement with the references. For the karate image, both outputs omit the board-breaking action and ViT misidentifies the participants' genders. On the roof-repair image, ViT's count of three people matches the references more closely than the CNN caption. These selected examples are illustrative, not a human evaluation or a representative estimate of error frequency.

TABLE IV
Computational efficiency comparison

Property	ResNet-50	ViT-B/16
Total parameters	35.11 M	96.72 M
Training epochs	22	22
Best val. loss	2.7038 (epoch 17)	2.6849 (epoch 17)
Total training time	188.50 min	492.27 min
Avg. end-to-end epoch time	8.57 min	22.38 min
Effective throughput*	≈56.4 img/s	≈21.6 img/s
Avg. latency / image	36.01 ms	37.55 ms



(a) ResNet-50 (b) ViT-B/16

Fig. 2. Training and validation cross-entropy by epoch for the final seed-42 runs. ViT validation loss reaches its minimum of 2.6849 at epoch 17.



Dog running. Reference: "A black and white dog is running through the grass." CNN generated: "a dog jumps through the air to catch a ball." ViT generated: "a black and white dog is running through the grass."



Karate board breaking. Reference: "A girl in karate uniform breaking a stick with a front kick." CNN generated: "two people are competing in a martial arts competition." ViT generated: "two men are practicing martial arts."



Roof repair. Reference: "Three men, one on a ladder, work on a roof." CNN generated: "two men are working on a rooftop." ViT generated: "three people are working on a roof."

Fig. 3. One reference caption and the greedy model-generated captions for three Flickr30k test images. All text is taken from the evaluation export. The first example shows an unsupported object hallucination by CNN; the second shows action and attribute errors; the third illustrates a difference in object counting.

VII. Discussion

A. Caption Quality

In this token-sequence experiment, ViT-B/16 has higher point estimates on all seven automatic metrics, with the largest absolute differences on CIDEr (+0.0232) and BLEU-4 (+0.0152). The paired bootstrap intervals exclude zero for BLEU-3 and BLEU-4, while intervals for BLEU-1, BLEU-2, METEOR, ROUGE-L, and CIDEr include zero. These are unadjusted test-set intervals for fixed checkpoints; they neither measure seed-to-seed variation nor justify a broad quality-superiority claim. Multi-seed training remains necessary to assess whether the observed differences are robust.

B. Computational Trade-offs

ViT-B/16 has about 2.76× as many parameters as the ResNet-50 system (96.72M vs. 35.11M). Its final training run took 2.61× as long (492.27 vs. 188.50 minutes), while its measured greedy latency was 1.04× higher (ViT: 37.55; ResNet: 36.01 ms/image). Both latency values were measured on the same Tesla T4 evaluation environment. Peak memory was not captured, so parameter count and wall-clock measurements should not be taken as a direct memory or deployment-footprint measurement.

C. Data Efficiency and Inductive Bias

Flickr30k supplies 29,000 training images in the selected split. The CNN grid and ViT patch sequence preserve spatially indexed features for cross-attention, but this experiment does not isolate inductive bias from the pretrained encoders, token count, optimization, or compute. The validation and test point estimates favor ViT on this run; no conclusion about generalization to larger or out-of-domain datasets follows without additional experiments.

D. Practical and Societal Considerations

Image captioning can support image search and assistive access for people who are blind or have low vision. However, an inaccurate caption can mislead a listener; the examples in Fig. 3 include an unsupported object and errors about an action or attribute. We did not conduct human assessment with intended users, so these results do not demonstrate accessibility, factual reliability, or deployment readiness. Future evaluation should measure correctness, relevance, and usefulness with human raters and diverse images.

E. Implications for Deployment

For this implementation, ResNet-50 is faster to fine-tune and has fewer parameters, while ViT-B/16 obtains higher automatic caption scores at a higher training and inference cost. Selecting

between them therefore depends on the deployment's quality, compute, and latency requirements. The single T4 result is not sufficient to generalize to other hardware or real-world image distributions.

VIII. Conclusion and Future Work

This paper compared ResNet-50 grid tokens and ViT-B/16 patch tokens with a shared Transformer decoder. On one seed and the Flickr30k Karpathy split, ViT had higher metric point estimates, while ResNet trained and decoded faster with fewer parameters. For these fixed checkpoints, unadjusted test-image bootstrap intervals exclude zero for BLEU-3 and BLEU-4; one seed does not establish run-to-run robustness, human preference, factual accuracy, or generalization. Future work should use multiple seeds, account for both seed and image uncertainty and multiple metric comparisons, conduct human and factuality evaluation, measure peak memory, and test additional datasets. A clearly identified BLIP-2 baseline is another useful comparison [19], [21].

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Appendix A
Code and Reproducibility Information

Source code, checkpoints, and manuscript: https://github.com/PradipPokhrel55/caption_generator.

Kaggle runs: HPO https://www.kaggle.com/code/pradippokhrel77/vit-hpo; CNN training https://www.kaggle.com/code/pradippokhrel45/cnn-transformer; ViT training https://www.kaggle.com/code/pradippokhrel45/vit-traning; evaluation/bootstrap https://www.kaggle.com/code/pradippokhrel45/evaluation-caption-generation.